

PRACTICAL – 4

'''

Implement a solution for a Constraint Satisfaction Problem using Branch and Bound and Backtracking for n-queens problem or a graph coloring problem.

'''

#Number of queens

```
print("Enter the number of queens: ")
```

```
N = int(input())
```

#chessboard

#NxN matrix with all elements 0

```
board = [[0] * N for _ in range(N)]
```

```
def is_attack(i, j):
```

```
    #checking if there is a queen in row or column
```

```
    for k in range(0, N):
```

```
        if board[i][k] == 1 or board[k][j] == 1:
```

```
            return True
```

```
    for k in range(0, N):
```

```
        for l in range(0, N):
```

```
            if (k + l == i + j) or (k - l == i - j):
```

```
                if board[k][l] == 1:
```

```
                    return True
```

```
    return False
```

```
def N_queen(n):
```

```
    if n == 0:
```

```
        return True
```

```
    for i in range(0, N):
```

```
        for j in range(0, N):
```

```
            if (not (is_attack(i, j))) and (board[i][j] != 1):
```

```
                board[i][j] = 1
```

```
                if N_queen(n - 1) == True:
```

```
                    return True
```

```
                board[i][j] = 0
```

```
    return False
```

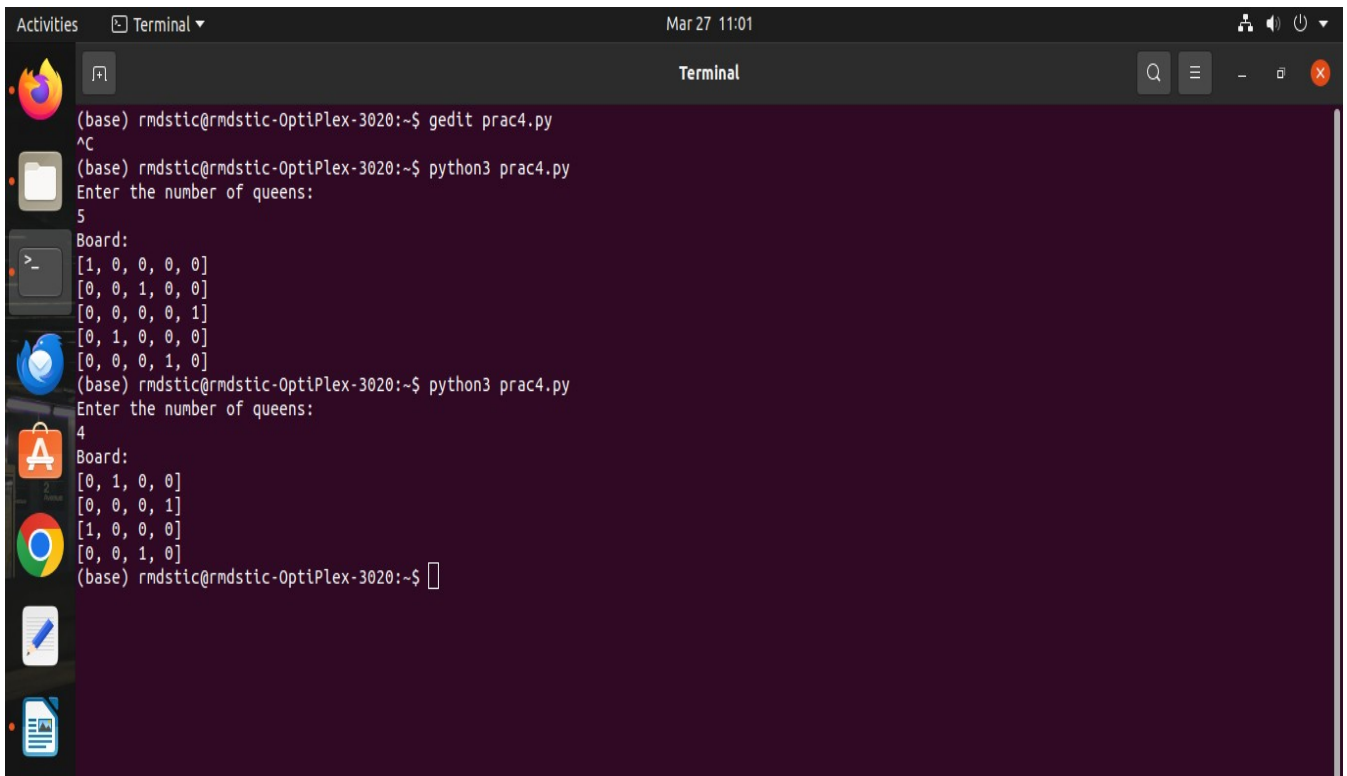
```
N_queen(N)
```

```
print('Board:')
```

```
for i in board:
```

```
    print(i)
```

OUTPUT



The image shows a terminal window titled "Terminal" with a dark background. The window displays the execution of a Python script named "prac4.py". The user enters the number of queens, and the program outputs a board configuration for 5 queens. The board is represented as a list of lists, where each inner list corresponds to a row and each element represents the column position of a queen. The output shows a valid solution for 5 queens.

```
(base) rmdstic@rmdstic-OptiPlex-3020:~$ gedit prac4.py
^C
(base) rmdstic@rmdstic-OptiPlex-3020:~$ python3 prac4.py
Enter the number of queens:
5
Board:
[1, 0, 0, 0, 0]
[0, 0, 1, 0, 0]
[0, 0, 0, 0, 1]
[0, 1, 0, 0, 0]
[0, 0, 0, 1, 0]
(base) rmdstic@rmdstic-OptiPlex-3020:~$ python3 prac4.py
Enter the number of queens:
4
Board:
[0, 1, 0, 0]
[0, 0, 0, 1]
[1, 0, 0, 0]
[0, 0, 1, 0]
(base) rmdstic@rmdstic-OptiPlex-3020:~$
```